

Object Thinking

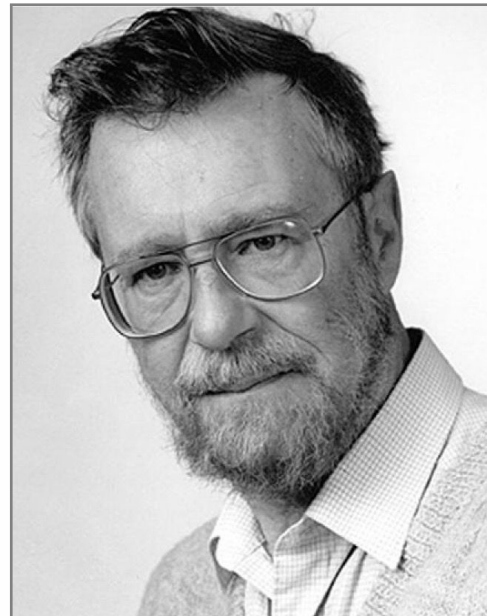
and Domain Driven Design

Yegor Bugayenko

Lecture #5 out of 16
80 minutes

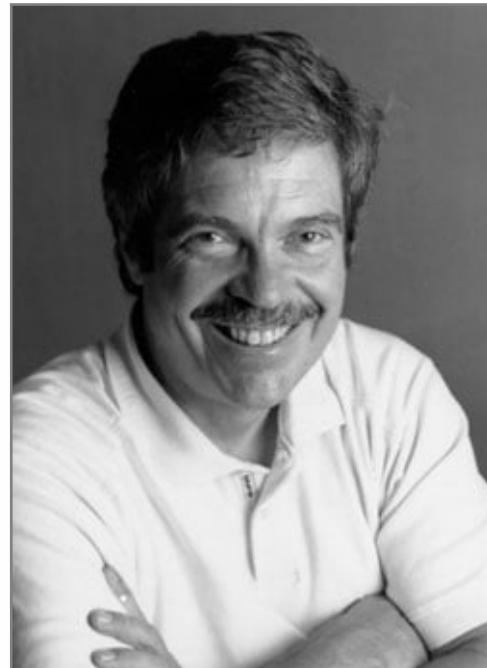
The slidedeck was presented by the author in this [YouTube Video](#).

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“Object-oriented programs are offered as alternatives to correct ones”

— Edsger W. Dijkstra (1989)



“I invented the term **object-oriented**, and I can tell you I did not have C++ in mind”

— Alan Kay (1997)



“Object-oriented programming offers a sustainable way to write spaghetti code”

— Paul Graham (2003)



“C++ is a horrible language. C++ leads to really, really bad design choices. ... idiotic object model crap.”

— Linus Torvalds (2007)



The Philosophy of OOP

What is an Object?

Three Most Evil Parts of OOP

Domain Driven Design

Elegant Objects

Books, Venues, Call-to-Action

Chapter #1:

The Philosophy of OOP

[[GOTO](#) IF/THEN CALL OOP₁ OOP₂]

The Era of GOTO

```
10 N = INT(RND(1) * 100)
20 T = T + 1
30 IF T > 5 THEN GOTO 120
40 PRINT "Guess a number in 0..99 range: "
50 INPUT X
60 IF X < N THEN PRINT "Too small."
70 IF X > N THEN PRINT "Too big."
80 IF X = N THEN GOTO 100
90 GOTO 20
100 PRINT "Bingo!"
110 GOTO 130
120 PRINT "You lost, sorry. It was: ", N
130 PRINT "Thanks for playing with me!"
```

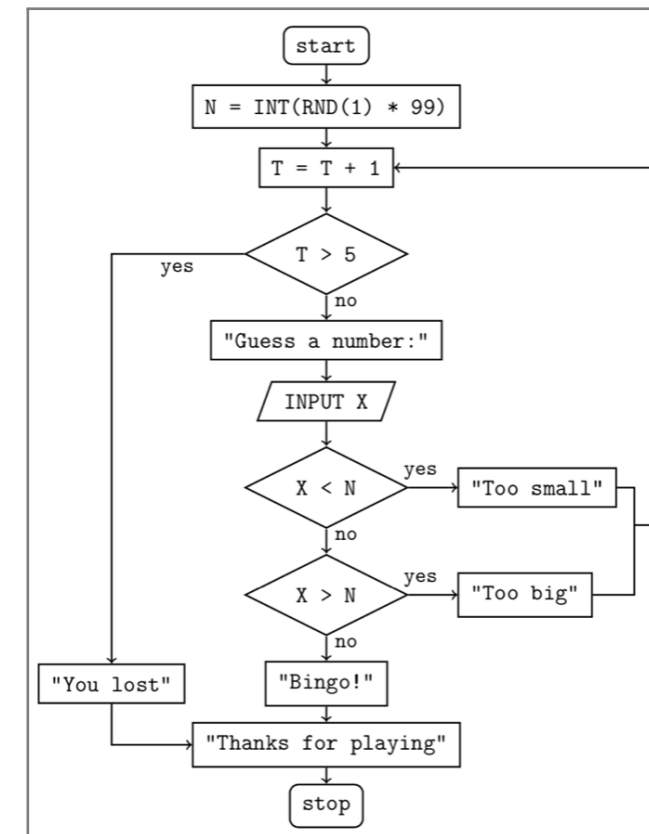
[GOTO IF/THEN CALL OOP₁ OOP₂]

Structured Programming

```
Uses sysutils;  
Var  
  N, X, T: Integer;  
Begin  
  Randomize;  
  N := Random(100);  
  T := 0;  
  While True Do Begin  
    T := T + 1;  
    If (T > 5) Then Begin  
      Writeln('You lost, sorry. It was: ' + IntToStr(N));  
      Break;  
    End;  
    Write('Guess a number in 0..99 range: ');  
    ReadLn(X);  
    If (X = N) Then Begin  
      Writeln('Bingo!');  
      Break;  
    End;  
    If X < N Then  
      Writeln('Too small.');
```

```
    If X > N Then  
      Writeln('Too big.');
```

```
  End;  
  Writeln('Thanks for playing with me!');  
End.
```



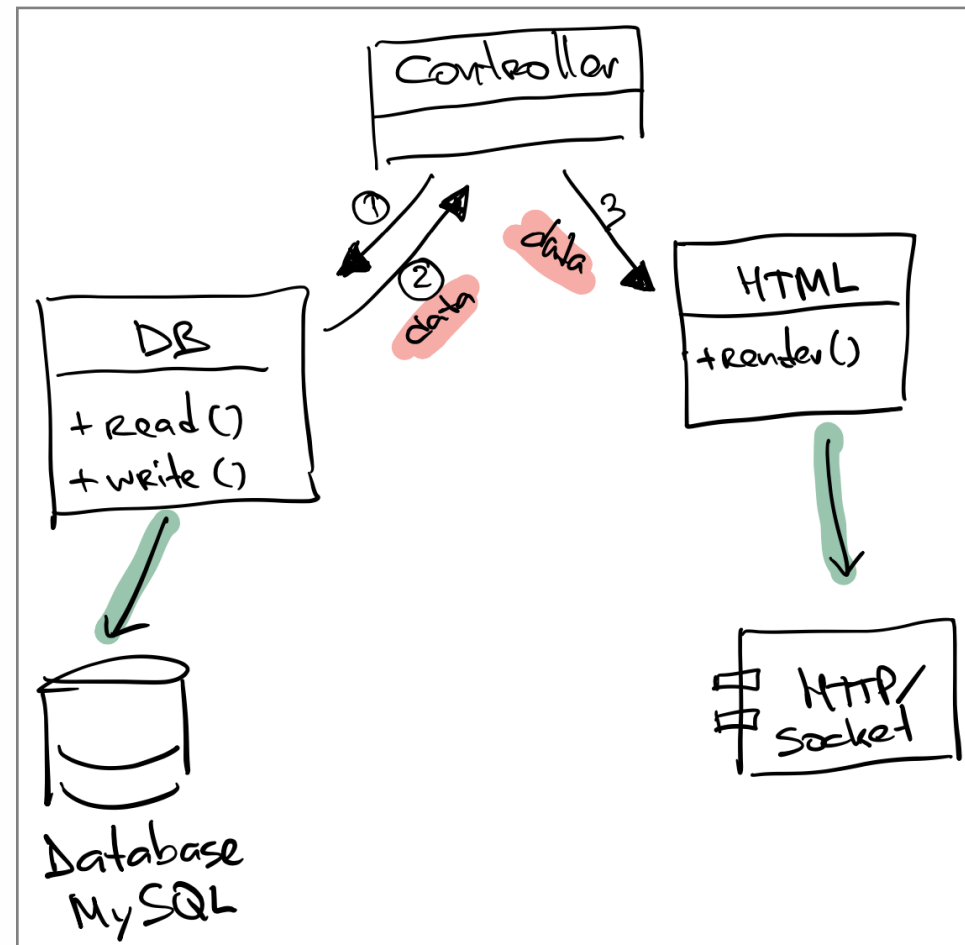
[GOTO IF/THEN [CALL](#) OOP₁ OOP₂]

Procedural Programming

```
atomic_long_set(&acct->count, 1);
init_fs_pin(&acct->pin, acct_pin_kill);
acct->file = file;
acct->needcheck = jiffies;
acct->ns = ns;
mutex_init(&acct->lock);
INIT_WORK(&acct->work, close_work);
init_completion(&acct->done);
mutex_lock_nested(&acct->lock, 1);
pin_insert(&acct->pin, mnt);
rcu_read_lock();
old = xchg(&ns->bacct, &acct->pin);
mutex_unlock(&acct->lock);
pin_kill(old);
mnt_drop_write(mnt);
mntput(mnt);
```

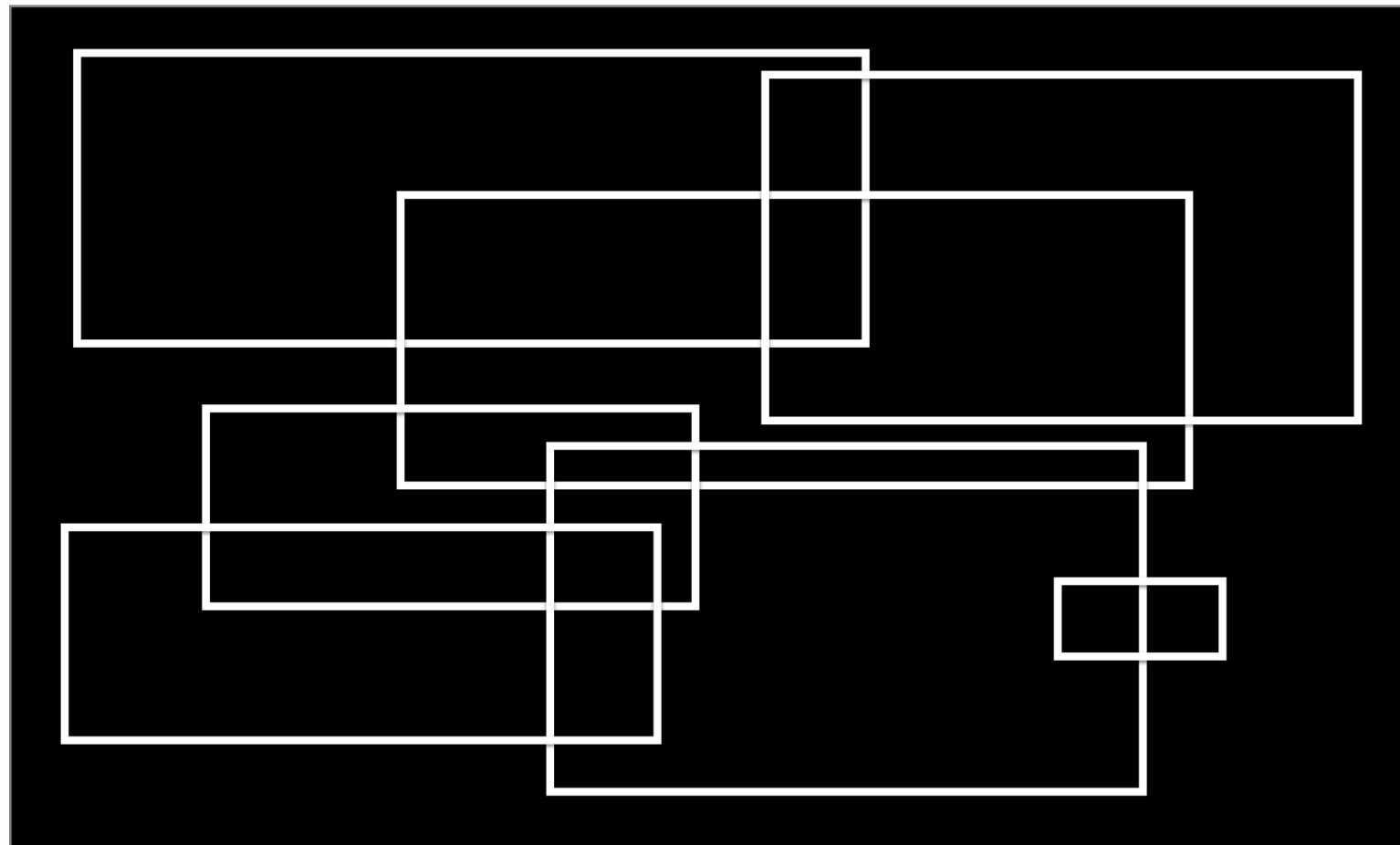
[GOTO IF/THEN CALL OOP₁ OOP₂]

Object-Oriented Programming ... Not!



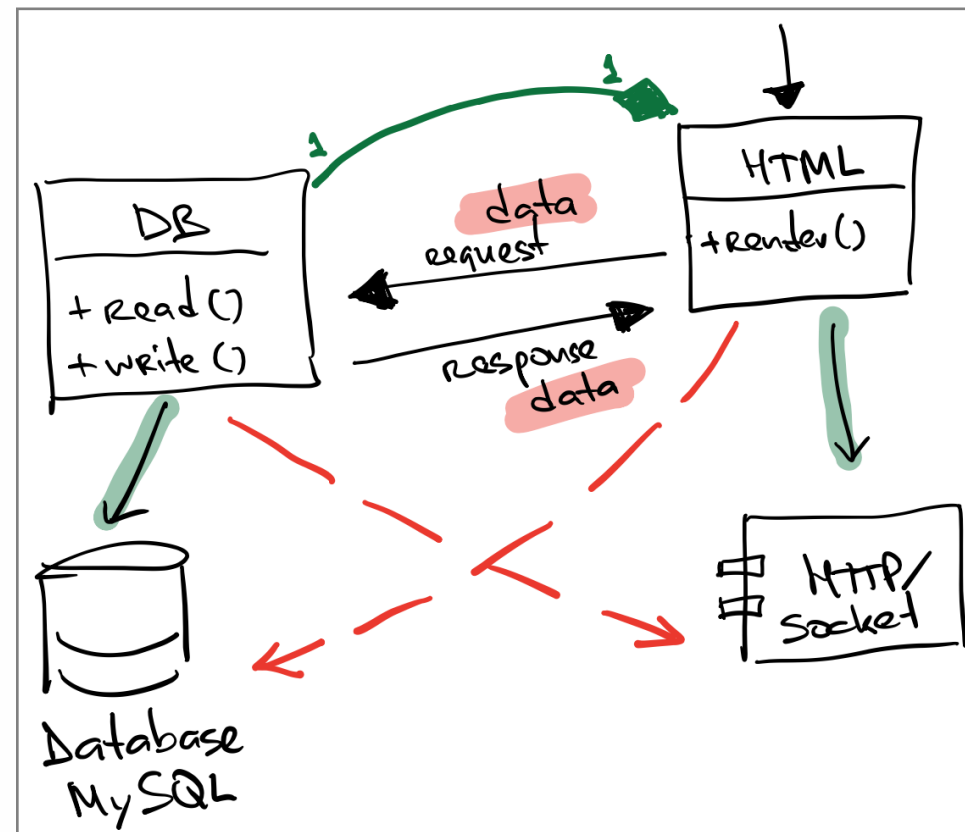
[GOTO IF/THEN CALL OOP₁ OOP₂]

Spaghetti OOP Code



[GOTO IF/THEN CALL OOP₁ OOP₂]

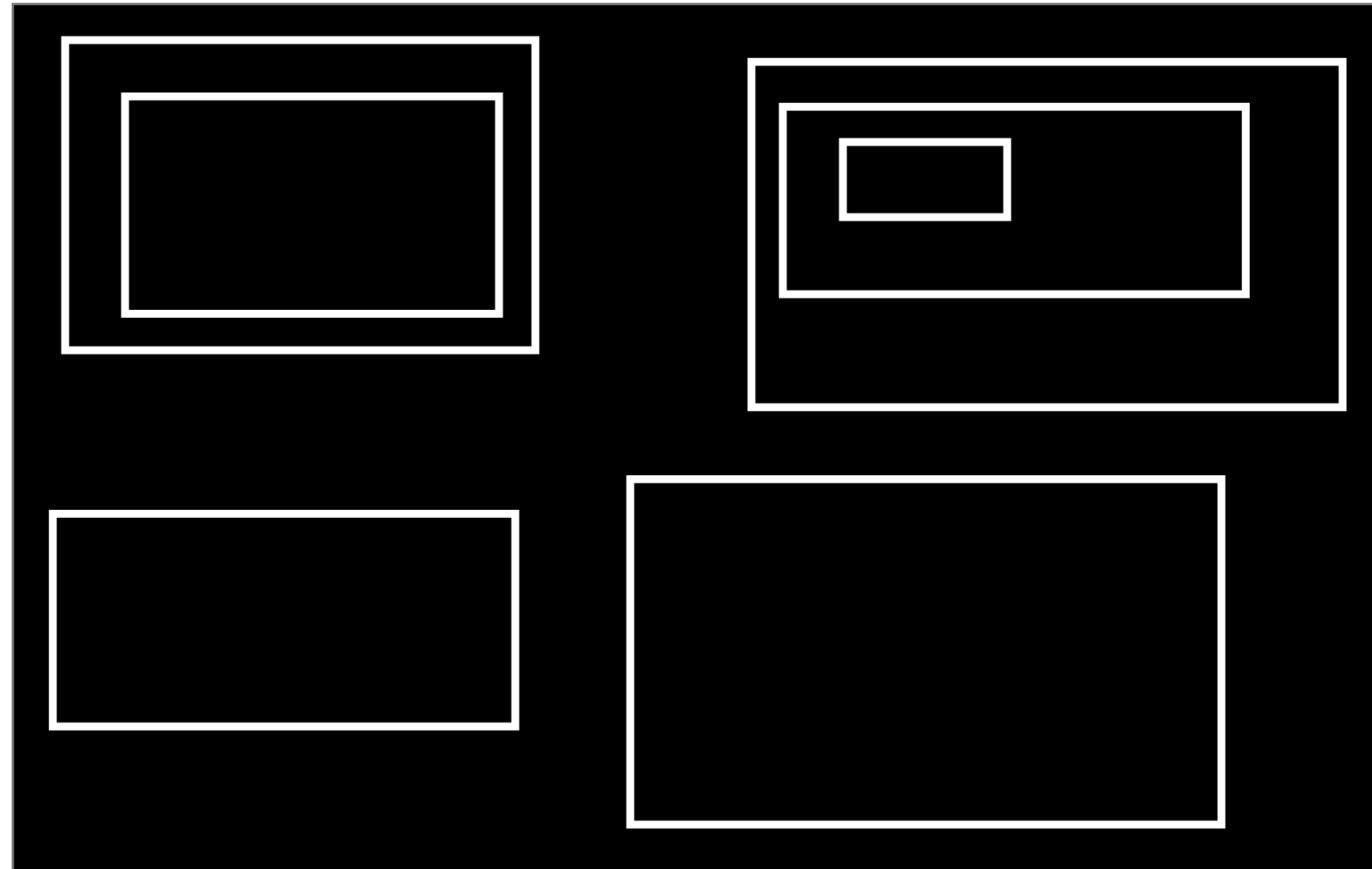
OOP Done Right



[GOTO IF/THEN CALL OOP₁ [OOP₂](#)]

Elegant OOP Code

[GOTO IF/THEN CALL OOP₁ OOP₂]



Chapter #2:

What is an Object?

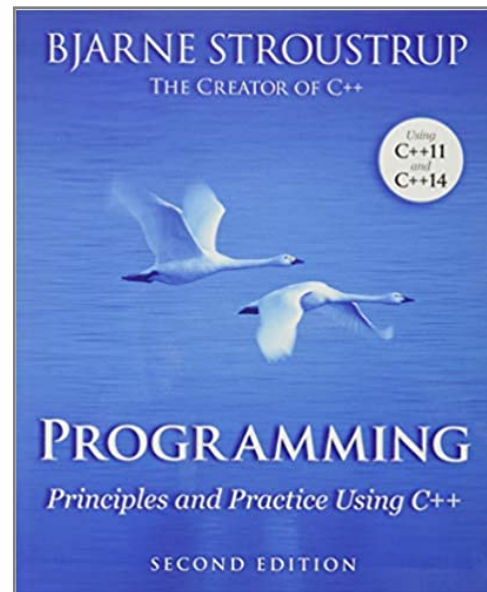
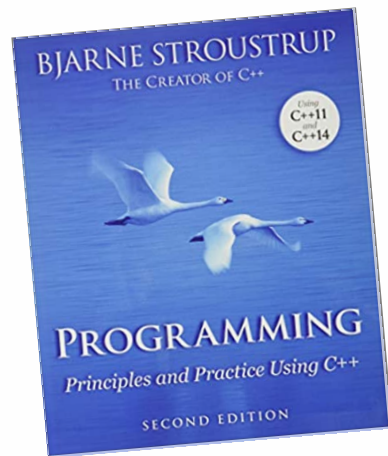
Theory [Object](#) Evil DDD Elegant Objects B.V.C.

17/50

[[C++](#) Wiki Smalltalk Java Eckel West]



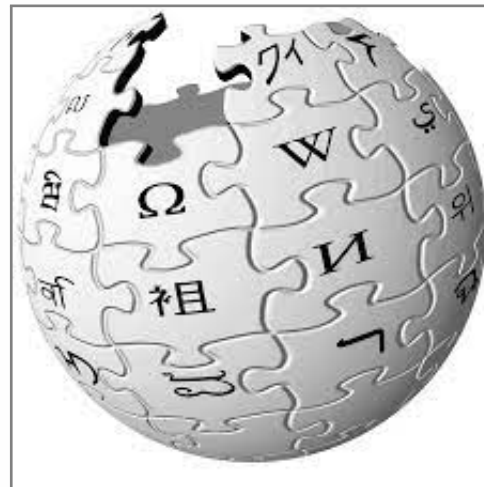
[C++ Wiki Smalltalk Java Eckel West]



“An object is some memory that holds a value of some type”

— Bjarne Stroustrup. *Programming: Principles and Practice Using C++*. 2nd ed. Pearson Education, 2014

Wiki



“Objects may contain data, in the form of fields, often known as attributes; and code, in the form of procedures, often known as methods”

— Wikipedia

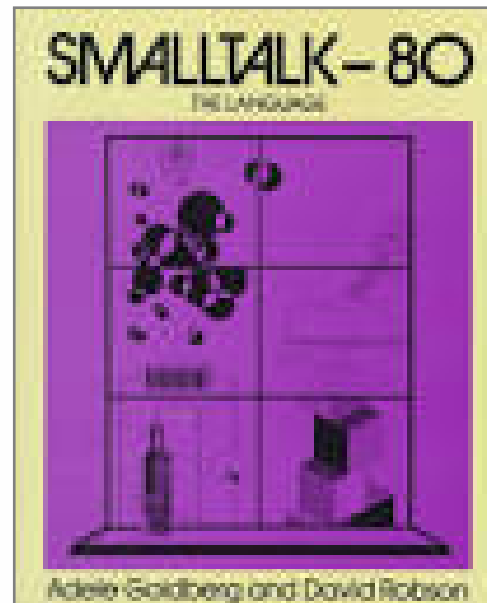
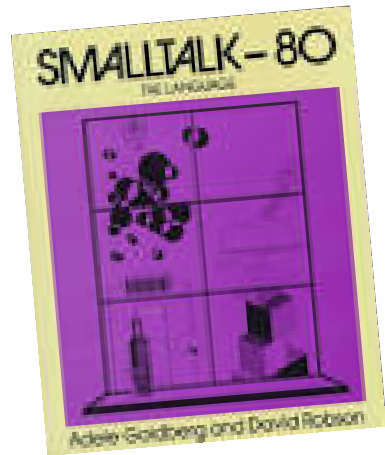
Theory [Object](#) Evil DDD Elegant Objects B.V.C.

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[C++ Wiki [Smalltalk](#) Java Eckel West]

Smalltalk

[C++ Wiki [Smalltalk](#) Java Eckel West]



“An object consists of some private memory and a set of operations”

— Adele Goldberg and David Robson. *Smalltalk-80: The Language and Its Implementation*. Addison-Wesley, 1983

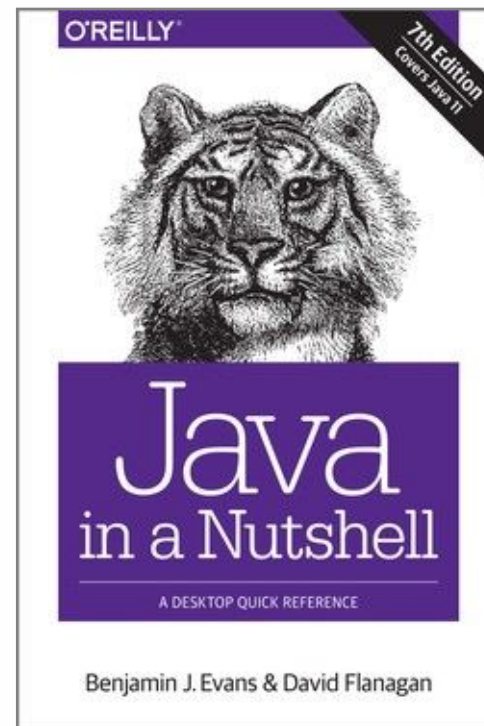
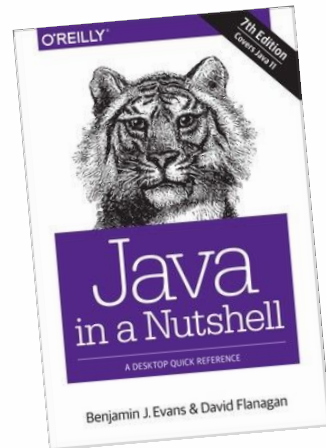
Theory [Object](#) Evil DDD Elegant Objects B.V.C.

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[C++ Wiki Smalltalk [Java](#) Eckel West]

Java

[C++ Wiki Smalltalk [Java](#) Eckel West]



“A class is a collection of data fields that hold values and methods that operate on those values”

— Benjamin J. Evans and David Flanagan. *Java in a Nutshell: A Desktop Quick Reference*. 7th ed. O'Reilly Media, 2018

Theory [Object](#) Evil DDD Elegant Objects B.V.C.

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[C++ Wiki Smalltalk Java [Eckel](#) West]

Eckel

[C++ Wiki Smalltalk Java [Eckel](#) West]



“Each object looks quite a bit like a little computer — it has a state, and it has operations that you can ask it to perform”

— Bruce Eckel. *Thinking in Java*. Prentice Hall, 2006. DOI: [10.5555/1076545](https://doi.org/10.5555/1076545)

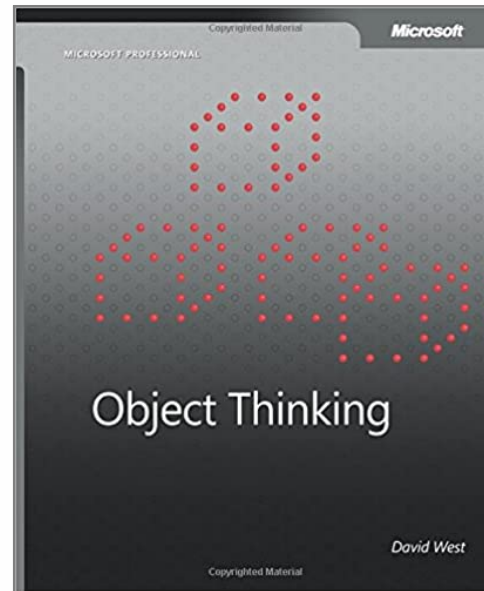
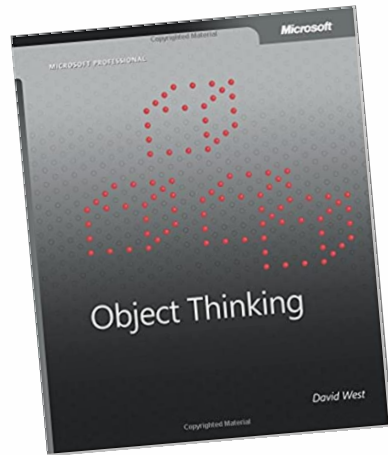
Theory [Object](#) Evil DDD Elegant Objects B.V.C.

26/50

[C++ Wiki Smalltalk Java Eckel [West](#)]

West

[C++ Wiki Smalltalk Java Eckel [West](#)]



“An object is the equivalent of the quanta from which the universe is constructed”

— David West. *Object Thinking*. Pearson Education, 2004. DOI: [10.5555/984130](https://doi.org/10.5555/984130)

Chapter #3:

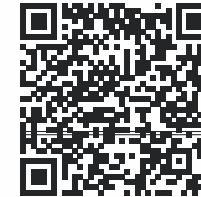
Three Most Evil Parts of OOP

[Static Mutability NULL]

1.

```
void transform(File in, File out) {  
    Collection<String> src = FileUtils.readLines(in, "UTF-8");  
    Collection<String> dest = new ArrayList<>(src.size());  
    for (String line : src) {  
        dest.add(line.trim());  
    }  
    FileUtils.writeLines(out, dest, "UTF-8");  
}
```

```
void transform(File in, File out) {  
    Collection<String> src = new Trimmed(  
        new FileLines(new UnicodeFile(in))  
    );  
    Collection<String> dest = new FileLines(  
        new UnicodeFile(out)  
    );  
    dest.addAll(src);  
}
```



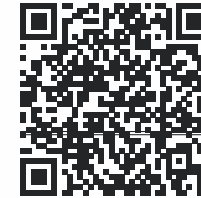
<https://www.yegor256.com/2014/05/05/oop-alternative-to-utility-classes.html> →

[Static Mutability NULL]

2. Mutability vs Immutability

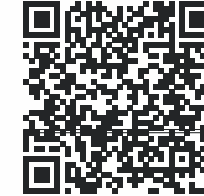
```
Email email = new SimpleEmail();
email.setHostName("smtp.googlemail.com");
email.setSmtpport(465);
email.setAuthenticator(new DefaultAuthenticator("user", "pwd"));
email.setFrom("yegor256@gmail.com", "Yegor Bugayenko");
email.addTo("dude@jcabi.com");
email.setSubject("how are you?");
email.setMsg("Dude, how are you?");
email.send();
```

```
Postman postman = new Postman.Default(
    new SMTP("smtp.googlemail.com", 465, "user", "pwd")
);
Envelope envelope = new Envelope.MIME(
    new Array<Stamp>(
        new StSender("Yegor Bugayenko <yegor256@gmail.com>"),
        new StRecipient("dude@jcabi.com"),
        new StSubject("how are you?")
    ),
    new Array<Enclosure>(
        new EnPlain("Dude, how are you?")
    )
);
postman.send(envelope);
```



<https://www.yegor256.com/2014/11/07/how-immutability-helps.html> →

[Static Mutability NULL]

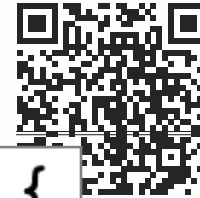


<https://www.yegor256.com/2014/06/09/objects-should-be-immutable.html>
⇒

Benefits of Immutability

- immutable objects are simpler to construct, test, and use
- truly immutable objects are always thread-safe
- they help to avoid temporal coupling
- their usage is side-effect free (no defensive copies)
- identity mutability problem is avoided
- they always have failure atomicity
- they are much easier to cache
- they prevent NULL references, which are bad

[Static Mutability [NULL](#)]



```
3. public Employee getByName(String name) {  
    int id = database.find(name);  
    if (id == 0) {  
        return null;  
    }  
    return new Employee(id);  
}
```

w.yegor256.com/2013/why-null-is-bad.html →

Null References, The Billion Dollar Mistake
presentation by Tony Hoare, [watch here](#).

[Static Mutability [NULL](#)]

NULL Object

```
public Employee getByName(String name) {  
    int id = database.find(name);  
    if (id == 0) {  
        return Employee.NOBODY;  
    }  
    return Employee(id);  
}
```



<https://www.yegor256.com/2015/08/25/fail-fast.html> →

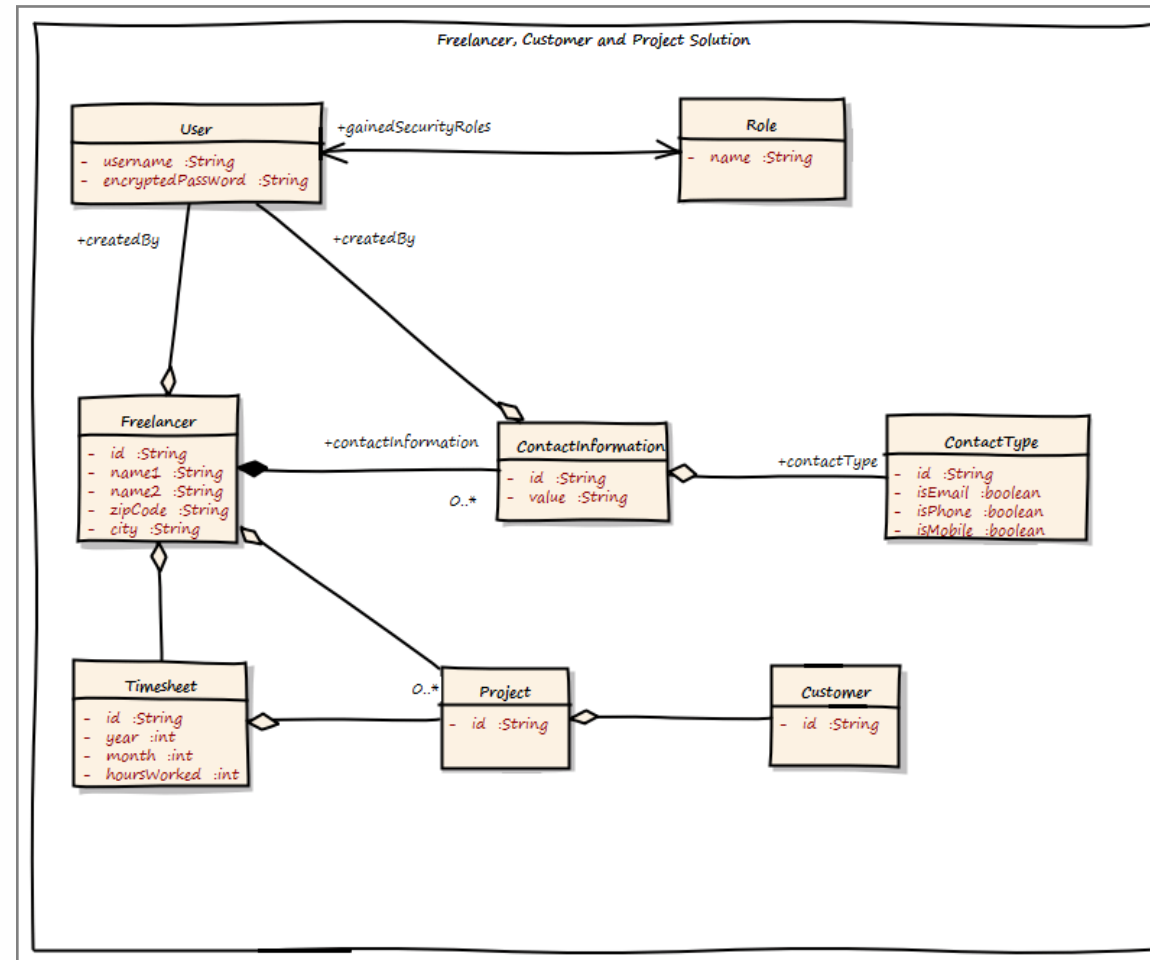
Fail Fast vs. Fail Safe

```
public Employee getByName(String name) {  
    int id = database.find(name);  
    if (id == 0) {  
        throw new EmployeeNotFoundException(name);  
    }  
    return Employee(id);  
}
```

Chapter #4:

Domain Driven Design

Names of Objects Done Right



Chapter #5:

Elegant Objects

Elegant Objects (EO)

Started in 2014

Two books, 40+ speeches, 80+ blog posts

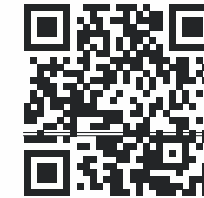
30+ frameworks and libraries

50+ fans registered

Six bloggers

e.g. pragmaticobjects.com, g4s8.wtf, amihaiemil.com

Five “Object Thinking” Meetups



<https://www.elegantobjects.org> ➞

Object-Oriented Lies

JPoint Student Day
Moscow, Russia, 22-24 April 2016



Java vs. OOP

JavaDay 2016
Minsk, Belarus, 11 June 2016



Java vs. OOP

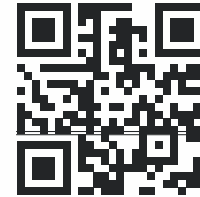
JavaDay Kyiv 2016
Kyiv, Ukraine, 15 October 2016



What's Wrong With OOP?

RigaDevDays 2017
Riga, Latvia, 15 May 2017





<https://www.eolang.org>
⇒

EOLANG: Our New Programming Language

```
[ ] > calculatesFibonacciNumberWithTail
eq. > @
  13
  fibonacci 7
[ n ] > fibonacci
  if. > @
    n.less 3
    small n
    rec n 1 1
[ n ] > small
  if. > @
    n.eq 2
    1
    n
[ n minus1 minus2 ] > rec
  if. > @
    n.eq 3
    minus1.add minus2
    rec (n.sub 1) (minus1.add minus2) minus1
```

If you want to help:

EO to JavaScript/Go/Rust/Ruby compilers

REPL for EO

Static analysis of EO code

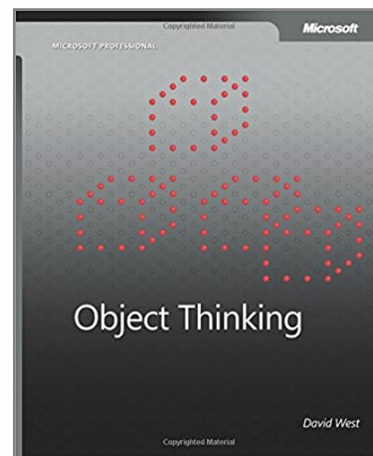
EO integration with Java/C++

Automated refactoring of EO code

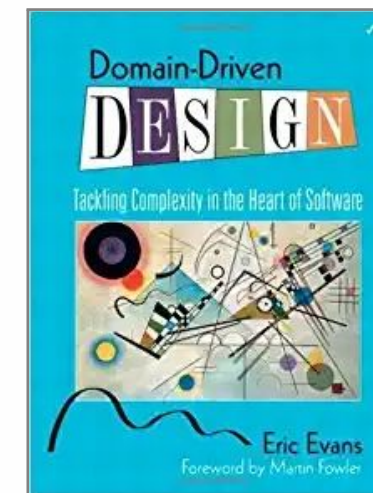
JetBrains plugin for EO

Chapter #6:

Books, Venues, Call-to-Action



David West. *Object Thinking*. Pearson Education, 2004. DOI: [10.5555/984130](https://doi.org/10.5555/984130)



Eric Evans. *Domain-Driven Design: Tackling Complexity in the Heart of Software*. Addison-Wesley, 2004. DOI: [10.5555/861502](https://doi.org/10.5555/861502)



Yegor Bugayenko. *Elegant Objects*. Amazon, 2016



Yegor Bugayenko. *Elegant Objects*. Amazon, 2017

Where to publish:

SPLASH: ACM SIGPLAN conference on Systems, Programming, Languages, and Applications

PLDI: ACM SIGPLAN Conference on Programming Language Design and Implementation

POPL: The Annual Symposium on Principles of Programming Languages

Call to Action:

Take `yegor256/hangman` repository and rewrite it in true object-oriented manner, submit a pull request with your changes.

Still unresolved issues:

- How to motivate coders for better OO practices?
- How to create better OO programming languages?
- How to catch bad OO practices automatically?
- How to prove some OO practices are bad?

References

- [1] Yegor Bugayenko. *Elegant Objects*. Amazon, 2016.
- [2] Yegor Bugayenko. *Elegant Objects*. Amazon, 2017.
- [3] Bruce Eckel. *Thinking in Java*. Prentice Hall, 2006. DOI: [10.5555/1076545](#).
- [4] Benjamin J. Evans and David Flanagan. *Java in a Nutshell: A Desktop Quick Reference*. 7th ed. O'Reilly Media, 2018.
- [5] Eric Evans. *Domain-Driven Design: Tackling Complexity in the Heart of Software*. Addison-Wesley, 2004. DOI: [10.5555/861502](#).
- [6] Adele Goldberg and David Robson. *Smalltalk-80: The Language and Its Implementation*. Addison-Wesley, 1983.
- [7] Bjarne Stroustrup. *Programming: Principles and Practice Using C++*. 2nd ed. Pearson Education, 2014.
- [8] David West. *Object Thinking*. Pearson Education, 2004. DOI: [10.5555/984130](#).